

MATH1860J Recitation Class 01

Foundations: Chapters 01-05

Shuhan Yu

SJTU Global College

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Outline

1. Introduction
2. Logic
3. Set Theory
4. Natural Numbers and Mathematical Induction
5. Rational Numbers
6. Real Numbers

1. Introduction

Introduction

About me:

- **Shuhan Yu:** Sophomore student majoring in ECE at GC.
- **Email:** barca0927@sjtu.edu.cn
- **Office Hours:** Every Monday, 20:20-22:20, LBL 326E.
- **Interest:** Machine Learning, optical communication and geoscience.

About Honors Mathematics Sequence (8 series):

- Emphasize on rigorous mathematical reasoning and proof.
- Write every step of your solution clearly and logically.
- Don't be frightened by "Honor" in the course title. Remember that 5 series is also an honor series.
- Sample exams are of great help. Go through them before exams.

Feel free to interrupt and ask questions if you have any doubt during RCs!

2. Logic

Definition

A **statement** (also called a **proposition**) is a declarative sentence that can be unambiguously classified as either *true* or *false*.

- **Logical operations:**

- Negation (unary operation): $\neg A$
- Conjunction: $A \wedge B$
- Disjunction: $A \vee B$
- Implication: $A \rightarrow B$
- Converse: $B \rightarrow A$
- Equivalence: $A \leftrightarrow B$

Binary operations return a compound statement.

- **Special propositions:**

- Tautology: always true
- Contradiction: always false
- Vacuous truth.

Propositional Logic

The definition of logical operations is often given by a **truth table**. Consider the implication $A \rightarrow B$. It can be rewritten in several equivalent forms (where $\equiv$ denotes logical equivalence):

$$A \rightarrow B \equiv \neg A \vee B \equiv \neg(A \wedge \neg B) \equiv \neg B \rightarrow \neg A.$$

A	B	$A \rightarrow B$	$\neg A \vee B$	$\neg(A \wedge \neg B)$	$\neg B \rightarrow \neg A$
T	T	T	T	T	T
T	F	F	F	F	F
F	T	T	T	T	T
F	F	T	T	T	T

With the basic operation laws established, we can use them to simplify and analyze more complicated propositions.

Propositional Logic

Remark: (Important for Proof)

- $A \rightarrow B \equiv \neg B \rightarrow \neg A$: equivalence with the **contrapositive**.
- $A \rightarrow B \equiv \neg(A \wedge \neg B)$: useful in **proof by contradiction**.
- $A \rightarrow B \equiv \neg A \vee B$: converts the implication into a disjunction, often used in algebraic manipulation (only using $\wedge, \vee, \neg$).

Important logical equivalences and laws:

- **Commutativity:** $p \vee q \equiv q \vee p$, $p \wedge q \equiv q \wedge p$
- **Associativity:** $(p \vee q) \vee r \equiv p \vee (q \vee r)$
- **Distributivity:**

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

- **De Morgan's Laws:**

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Predicate Logic

A **statement** can be expressed as:

statement = **quantifier** + **predicate (statement frame)**.

Definition

A **predicate** is an expression $E(x)$ that becomes a statement when x is replaced by an object from a **specified class** of objects.

- **Universal quantifier** $\forall$: “for all”. The expression $\forall x \in X : E(x)$ means “Every object x in X satisfies property E ”.
- **Existential quantifier** $\exists$: “there exists”. The expression $\exists x \in X : E(x)$ means “There exists at least one object x in X with property E ”.
- **Unique existential** $\exists!$: exactly one such object exists.

Predicate Logic

In all cases, **the order of the quantifiers is significant**. Thus,

$$\forall x \exists y : E(x, y) \not\Leftrightarrow \exists y \forall x : E(x, y)$$

- Former, for all x there exists some y such that $E(x, y)$ is true.
- Latter, there exists a fixed y such that $E(x, y)$ is true for all x .

Homework Exercise 1.3:

$$\exists_{0 \in \mathbb{Q}} \forall_{a \in \mathbb{Q}} a + 0 = 0 + a = a$$

and

$$\forall_{a \in \mathbb{Q}} \exists_{0 \in \mathbb{Q}} a + 0 = 0 + a = a.$$

Exercise

Exercise 01

- 1 Show that:

$$A \Leftrightarrow B \equiv \neg((\neg A \wedge B) \vee (\neg B \wedge A))$$

- 2 Give the negations of the statements:

$$\forall x \in X, \exists y \in Y : E(x, y, z)$$

$$\forall \epsilon > 0, \exists \delta > 0 : |x - a| < \delta \implies |f(x) - f(a)| < \epsilon$$

Remarks:

Using the quantifiers $\exists$ and $\forall$, negation becomes a purely “mechanical” process in which the symbols $\exists$ and $\forall$ (as well as $\wedge$ and $\vee$) are interchanged (without changing the order), and statements are negated.

3. Set Theory

Set Basics and Operations

Sets are collections of objects, called **elements** or **members**.

- Define by Listing: $A = \{1, 2, 3, 4\}$
- Define by Predicate: $A = \{x \in \mathbb{N} \mid 1 \leq x \leq 4\}$
- Set $A = B$ iff $A \subseteq B$ and $B \subseteq A$. (Useful in Proof)

Let $A = \{x \mid P_1(x)\}$ and $B = \{x \mid P_2(x)\}$.

- Union: $A \cup B := \{x \mid P_1(x) \vee P_2(x)\}$
- Intersection: $A \cap B := \{x \mid P_1(x) \wedge P_2(x)\}$
- Difference: $A \setminus B := \{x \mid P_1(x) \wedge \neg P_2(x)\}$.
- Complement: $A^c(\overline{A}) := \mathbb{U} \setminus A := \{x \in \mathbb{U} \mid \neg P_1(x)\}$.

If $A \cap B = \emptyset$, we say that A and B are **disjoint**.

Remark: Distinguish between **Difference**($A \setminus B$) and **Quotient**(A/B)

Power Set and Cardinality

- **Power Set:** $P(A) = \{X \mid X \subseteq A\}$
- **Cardinality:** $|A|$ is the number of elements in set A
- **Finite Set:** A set with a finite number of elements
- **Infinite Set:** A set with an infinite number of elements
- **Countable Set:** A set that can be put into one-to-one correspondence with the natural numbers (e.g., $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$)
- **Uncountable Set:** A set that cannot be put into one-to-one correspondence with the natural numbers (e.g., $\mathbb{R}$)
- **Cardinality of Power Set:** $|P(A)| = 2^{|A|}$ if A is finite

Ordered Pairs and Cartesian Product

- **Ordered Pair:** (a, b)
- **Cartesian Product:** $A \times B = \{(a, b) \mid a \in A, b \in B\}$
- **n-tuple:** generalization of ordered pair, $(a_1, a_2, \dots, a_n)$
- Construction of ordered pair and natural numbers:
 $(a, b) = \{\{a\}, \{a, b\}\}; 0 = \{\emptyset\}, 1 = \{\emptyset\}, 2 = \{\emptyset, \{\emptyset\}\}, \dots$

- Note that $(a, b) = (c, d) \iff a = c \wedge b = d$.
- In general, $A \times B \neq B \times A$.

Exercise 02 (True or False)

Determine whether each statement is true or false. Justify your answer.

- ❶ For any sets A , B , and C , $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
- ❷ If A and B are finite sets, then $|A \times B| = |A| \cdot |B|$.
- ❸ For any sets A , B , and C , if $A \times C = B \times C$ and $C \neq \emptyset$, then $A = B$.
- ❹ For any set A , if $|A| = n$, then $|P(P(A))| = 2^{2^n}$.

Naive Set Theory and Russell's Paradox

Naive Set Theory: You can define a set by any property. A set is any collection of distinct objects. **Problem: Russell's Paradox** arises when considering the set of all sets that do not contain themselves:

$$S = \{x \mid x \text{ is a set and } x \notin x\}$$

Question: Is $S \in S$?

- If $S \in S$, then by definition $S \notin S$.
- If $S \notin S$, then by definition $S \in S$.

Conclusion: Contradiction arises. Not every property can define a set freely. $\Rightarrow$ **Implication:** Motivates axiomatic set theory (e.g., Zermelo-Fraenkel Set Theory, ZFC) to avoid such paradoxes.

Cantor's Paradox (Supplement)

Prove that “the set of all sets” does not exist.

Cantor's Theorem: For any set A ,

$$|A| < |P(A)|,$$

where $P(A)$ is the power set of A , i.e., the set of all subsets of A .

Consider: The “set of all sets” U .

- By definition, U contains every set, so $P(U) \subseteq U$.
- By Cantor's Theorem, $|U| < |P(U)|$.

Contradiction: $|U| \geq |P(U)|$ from definition, but $|U| < |P(U)|$ by Cantor.

Conclusion: There is no “set of all sets”. This prevents the paradox.

Zermelo-Fraenkel Set Theory (ZF)

- ➊ **Axiom of Extensionality:** Two sets are equal if they have the same elements.
- ➋ **Axiom of Empty Set:** There exists a set with no elements, denoted $\emptyset$.
- ➌ **Axiom of Pairing:** For any sets a and b , there exists a set $\{a, b\}$.
- ➍ **Axiom of Union:** For any set of sets A , there exists a set $\bigcup A$ containing all elements of elements of A .
- ➎ **Axiom of Power Set:** For any set A , there exists a set $P(A)$ of all subsets of A .
- ➏ **Axiom of Infinity:** There exists a set containing $\emptyset$ and closed under successor.
- ➐ **Axiom of Replacement:** The image of a set under any definable function is also a set.
- ➑ **Axiom of Separation (Specification):** For any set A and property P , $\{x \in A \mid P(x)\}$ is a set.
- ➒ **Axiom of Regularity (Foundation):** Every non-empty set A contains an element disjoint from A .

Exercise

Exercise 03

Let A, B, C be three statements. Use truth table to prove that

$$(A \vee B) \wedge C \equiv (A \wedge C) \vee (B \wedge C)$$

Let A, B, C be three sets. Prove that (may use Venn Diagram)

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

4. Natural Numbers and Mathematical Induction

Natural Numbers

Natural Numbers from ZF Set Theory

- By the **Axiom of Infinity**, there exists a set I such that:
 - $\emptyset \in I$
 - For each $x \in I$, $S(x) := x \cup \{x\} \in I$
- **Define** natural numbers **recursively**:

$$0 := \emptyset, \quad 1 := S(0) = \{\emptyset\}, \quad 2 := S(1) = \{\emptyset, \{\emptyset\}\}, \dots$$

- The set of natural numbers $\mathbb{N}$ is the smallest set containing 0 and closed under S .
- Ordering: The natural numbers are well-ordered.

Comment: $\{\cdot\}$ is packaging and $\cup$ is stacking.

Each new number encompasses all its previous history and incorporates “the previous self” as a new element. This is why natural numbers have sequentiality.

Recursive Definition of Addition on $\mathbb{N}$

- **Base case:**

$$\forall n \in \mathbb{N}, \quad n + 0 := n$$

- **Successor step:**

$$\forall n, m \in \mathbb{N}, \quad n + S(m) := S(n + m)$$

Explanation: $S(m) = m \cup \{m\}$ is the successor of m , so addition recursively “adds 1” using the successor operation.

Properties of addition:

- Commutativity: $n + m = m + n$
- Associativity: $(n + m) + k = n + (m + k)$
- Identity/Existence of a neutral element: $n + 0 = n$

Natural Numbers

Recursive Definition of Multiplication on $\mathbb{N}$

- **Base case:**

$$\forall n \in \mathbb{N}, \quad n \cdot 0 := 0$$

- **Successor step:**

$$\forall n, m \in \mathbb{N}, \quad n \cdot S(m) := (n \cdot m) + n$$

Explanation: Multiplication is defined as repeated addition, consistent with the intuitive meaning of natural numbers.

Properties of multiplication:

- Commutativity: $n \cdot m = m \cdot n$
- Associativity: $(n \cdot m) \cdot k = n \cdot (m \cdot k)$
- Identity/Existence of a neutral element: $n \cdot 1 = n$
- Distributivity: $n \cdot (m + k) = (n \cdot m) + (n \cdot k)$

Definition (Principle of Mathematical Induction)

Let $A(n)$ be a statement depending on $n \in \mathbb{N}$ and let $n_0 \in \mathbb{N}$. To prove $A(n)$ for all $n \geq n_0$, it suffices to verify:

- 1 **Base Case:** $A(n_0)$ is true.
- 2 **Inductive Step:** For all $n \geq n_0$, if $A(n)$ is true, then $A(n+1)$ is true:

$$\forall n \in \mathbb{N}, n \geq n_0 : A(n) \implies A(n+1).$$

Conclusion: If both conditions hold, $A(n)$ is true for all $n \geq n_0$.

Remark: The principle of induction is essentially part of the definition of the natural numbers $\mathbb{N}$.

Mathematical Induction

Exercise 04

Suppose M is a finite set of men.

Proposition: For any $a, b \in M$, $a = b$.

Proof (by induction on $|M|$):

- ① **Base Case:** If $|M| = 1$, the claim is trivially true.
- ② **Inductive Step:** Assume true for n men. Let $|M| = n + 1$ and $a, b \in M$. Define $M_a = M \setminus \{a\}$, $M_b = M \setminus \{b\}$. Take $c \in M_a \cap M_b$. Then $a = c, b = c \implies a = b$.

Question: Where is the flaw in this proof?

Exercise 05

Use mathematical induction, find the general term formula for the Fibonacci sequence defined recursively by:

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2.$$

5. Rational Numbers

Rational Numbers and Basic Properties

The set of rational numbers is introduced from set theory as

$$\mathbb{Q} := \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

Recap the basic properties of rational numbers:

Properties for Addition:

- Commutativity: $a + b = b + a$
- Associativity:
 $(a + b) + c = a + (b + c)$
- Existence of additive identity:
 $a + 0 = a$
- Existence of additive inverse:
 $a + (-a) = 0$

Properties for Multiplication:

- Commutativity: $a \cdot b = b \cdot a$
- Associativity:
 $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- Existence of multiplicative identity: $a \cdot 1 = a$
- Existence of multiplicative inverse: $a \cdot a^{-1} = 1$ for $a \neq 0$
- Distributivity:
 $a \cdot (b + c) = a \cdot b + a \cdot c$

Absolute Value and Triangle Inequality

Definition (Absolute Value)

The **absolute value** (also called the **modulus**) of a rational number x is defined as

$$|x| := \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Theorem (Triangle Inequality)

For any $x, y \in \mathbb{Q}$, the following inequality holds:

$$|x + y| \leq |x| + |y|.$$

Inverse Triangle Inequality:

$$||x| - |y|| \leq |x - y|.$$

Trichotomy Law and Closedness

We assume that we know what a **strictly positive** rational number is, i.e., that there exists a set P with the property that for every number $a \in \mathbb{Q}$, one and only one of the following holds:

- 1 $a \in P$ (i.e., a is strictly positive),
- 2 $-a \in P$ (i.e., a is strictly negative),
- 3 $a = 0$.

This is known as the **trichotomy law** for rational numbers (**P10**). It allows us to compare any two rational numbers and determine their order.

We also assume that the set of positive numbers is **closed** under addition and multiplication:

- 1 If $a, b \in P$, then $a + b \in P$. (**P11**)
- 2 If $a, b \in P$, then $a \cdot b \in P$. (**P12**)

6. Real Numbers

Bounds and Boundedness

Definition (Upper and Lower Bounds)

Let $(X, \leq)$ be a partially ordered set and $A \subseteq X$ be nonempty.

- An element $s \in X$ is an **upper bound** of A if $a \leq s$ for all $a \in A$.
- An element $s \in X$ is a **lower bound** of A if $a \geq s$ for all $a \in A$.

Definition (Bounded Sets in $\mathbb{Q}$)

A set $U \subset \mathbb{Q}$ is said to be **bounded** if there exists a constant $c \in \mathbb{Q}$ such that

$$|x| \leq c \text{ for all } x \in U.$$

If this is not the case, we say that U is **unbounded**.

We say that a set $U \subset \mathbb{Q}$ is **bounded above** if there exists an upper bound for U , and **bounded below** if there exists a lower bound for U .

Maximum and Minimum, Supremum and Infimum

Definition (Maximum and Minimum)

Let $U \subset \mathbb{Q}$.

- $x_1 \in U$ is the **minimum** of U ($x_1 = \min U$), if $x_1 \leq x$ for all $x \in U$.
- $x_2 \in U$ is the **maximum** of U ($x_2 = \max U$), if $x_2 \geq x$ for all $x \in U$.

Note: A set not bounded below has no minimum, and a set not bounded above has no maximum. Even a bounded set may not have a max or min.

Definition (Supremum and Infimum)

Let $U \subset \mathbb{Q}$.

- An upper bound $c_2 \in \mathbb{Q}$ of U is the **least upper bound** or **supremum**, denoted $c_2 = \sup U$, if no $c \in \mathbb{Q}$ with $c < c_2$ is an upper bound.
- A lower bound $c_1 \in \mathbb{Q}$ of U is the **greatest lower bound** or **infimum**, denoted $c_1 = \inf U$, if no $c \in \mathbb{Q}$ with $c > c_1$ is a lower bound.

Greatest Lower and Least Upper Bounds of Sets

Definition

We say that an upper bound $c_2 \in \mathbb{Q}$ of a set $U \subset \mathbb{Q}$ is the **least upper bound** or **supremum** of U if no $c \in \mathbb{Q}$ with $c < c_2$ is an upper bound. We then write $c_2 =: \sup U$.

We say that a lower bound $c_1 \in \mathbb{Q}$ is the **greatest lower bound** or **infimum** of U if no $c \in \mathbb{Q}$ with $c > c_1$ is a lower bound. We then write $c_1 =: \inf U$.

Naive Real Number

Problem: Rational numbers are insufficient.



We define the set of real numbers $\mathbb{R}$ as the smallest extension of the rational numbers $\mathbb{Q}$ such that the following property holds:

Least Upper Bound Property (P13)

If $A \subset \mathbb{R}$, $A \neq \emptyset$, and A is bounded above, then there exists a least upper bound for A in $\mathbb{R}$.

We call all real numbers that are not rational “**irrational numbers**”.

Exercise 06

Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

Subsets of the Real Numbers

Definition

Let $A \subset \mathbb{R}$ be any set.

- ❶ We call $x \in \mathbb{R}$ an **interior point** of A if there exists some $\varepsilon > 0$ such that the interval $(x - \varepsilon, x + \varepsilon) \subset A$. The set of all interior points of A is denoted by $\text{int } A$.
- ❷ We call $x \in \mathbb{R}$ an **exterior point** of A if there exists some $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \cap A = \emptyset$.
- ❸ We call $x \in \mathbb{R}$ a **boundary point** of A if for every $\varepsilon > 0$, both $(x - \varepsilon, x + \varepsilon) \cap A \neq \emptyset$ and $(x - \varepsilon, x + \varepsilon) \cap A^c \neq \emptyset$. The set of all boundary points of A is denoted by ∂A .
- ❹ We call $x \in \mathbb{R}$ an **accumulation point** of A if for every $\varepsilon > 0$, $(x - \varepsilon, x + \varepsilon) \cap (A \setminus \{x\}) \neq \emptyset$.

Subsets of the Real Numbers

Definition

- ❶ A set $A \subset \mathbb{R}$ is called **open** if all points of A are interior points; that is, if $A = \text{int } A$.
- ❷ A set $A \subset \mathbb{R}$ is called **closed** if its complement $\mathbb{R} \setminus A$ is open.
- ❸ The set

$$\overline{A} := A \cup \partial A$$

is called the **closure** of A . It can be shown that $\overline{A}$ is the smallest closed set that contains A .

Exercise 07

Show that $\mathbb{R} = \text{int } A \cup \text{ext } A \cup \partial A$, and think about the sets that are both open and closed.

Exercise 08

Proof that:

- (a) For any collection $\{G_\alpha\}$ of open sets, the union $\bigcup_\alpha G_\alpha$ is open.
- (b) For any collection $\{F_\alpha\}$ of closed sets, the intersection $\bigcap_\alpha F_\alpha$ is closed. (2.21)
- (c) For any finite collection $G_1, G_2, \dots, G_n$ of open sets, the intersection $\bigcap_{i=1}^n G_i$ is open.
- (d) For any finite collection $F_1, F_2, \dots, F_n$ of closed sets, the union $\bigcup_{i=1}^n F_i$ is closed.

Assignment Problems Q & A

- Format and English Usage
- Disjunctive Normal Form
- Truth Table
- Functionally Complete
- Assignment Group Discussion
- ...Interest in Cantor's Theorem?

References:

- Horst Hohberger, *math186_all_lecture_slides (2026)*
- Wenxin Chen, *MATH1860J RC*, 25 Fall
- Leqi Tang, *MATH1860J RC*, 25 Fall

To be continued...

- Complex Numbers
- Functions and Maps
- Sequences

Thanks for Listening!

Any questions? Please feel free to ask!